Halving Count Structure of Odd Integers in the Collatz Problem

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Abstract

We study the Collatz problem by first showing that proving the process for the odd numbers proves it for all numbers. We therefore restrict attention to the odd numbers and induce the map $f(x) = \frac{3x+1}{2^{n(x)+1}}$ where $n(x)$ counts the number of halving steps required to get to the next odd number. We show that the odd numbers split exactly into separate series, one for each value of n where every series is an arithmetic progression and no two share a term

The Collatz Conjecture

Few unsolved problems in mathematics can be explained to a child in under a minute. The Collatz conjecture is one of them. Take any positive integer. If it is even, halve it; if it is odd, multiply it by three and add one. Repeat. The conjecture claims that no matter which number you start from, you always eventually reach 1. We can write the rule as a single map:

$$C(x) = \begin{cases} x/2 & \text{if } x \text{ is even} \\ 3x+1 & \text{if } x \text{ is odd} \end{cases}$$

The rule is trivial to state and trivial to compute, yet whether it holds for *every* starting integer has resisted proof since Lothar Collatz first posed it in 1937 It has been checked by computer for all starting values up to 2^{71} without a single exception but still no proof exists. It is, in a common phrasing, the simplest problem no one can solve.

Reducing the problem to odd numbers

Before attacking the full map, we make a simplifying observation that will shape everything that follows. Every positive integer can be written uniquely as

$$N = 2^n \times \text{odd}$$

This gives us a genuine reduction, and so I state that:

If the Collatz conjecture holds for all odd numbers, it holds for all positive integers.

Formulating the Size

To prove the conjecture for an odd number x is to prove that the $Size$ of the series must be *finite*.

Every step from one odd number to the next odd number is a burst of halvings, and the count of those halvings is exactly the n we defined. If we sum this count over every step in the trajectory of a given starting odd number, we obtain the $Size$. This can be mathematically formulated as:

$$Size = \sum_{i \in S} n\big(f_i(x)\big), \qquad f_i(x) = \underbrace{f(f(\cdots f(x)\cdots))}_{i \text{ times}}$$

Where $n(x)$ is the function defining the number of steps in the halving burst between two odds, corresponding to the first odd x, $f(x)$ is a simple map that takes us from one odd number to the next one as per the rules of Collatz Conjecture and S is the set over which the summation is valid as per the rules of this problem.

Formulating $f(x)$ and S

A function for going from one odd number to the next one in a Collatz Series is very intuitive. As we discussed in the Third Section, any even number can be directly decomposed into an odd number by halving it n times and we also know that any odd number "x" can be converted into an even number by performing "$3x + 1$". So the map comes out to be:

$$f(x) = \frac{3x + 1}{2^{\,n(x)+1}}$$

The set S is also very intuitive, the summation can be performed only for $i \in \mathbb{W}$ where $f_i(x) \neq 1$ since the $1 \to 4 \to 2 \to 1$ loop starts and there is no point adding beyond that.
Thus, the set S comes out to be:

$$S = \{\, i : i \in \mathbb{W},\ f_i(x) \neq 1 \,\}$$

Formulating $n(x)$

The formulation of $n(x)$ i.e number of halvings possible from $\frac{3x+1}{2}$ is genuinely the most beautiful part about this solution. Upon computing values of n corresponding to every odd number x we got the table:

x	$n(x)$	x	$n(x)$	x	$n(x)$	x	$n(x)$	x	$n(x)$
1	1	27	0	53	4	79	0	105	1
3	0	29	2	55	0	81	1	107	0
5	3	31	0	57	1	83	0	109	2
7	0	33	1	59	0	85	7	111	0
9	1	35	0	61	2	87	0	113	1
11	0	37	3	63	0	89	1	115	0
13	2	39	0	65	1	91	0	117	4
15	0	41	1	67	0	93	2	119	0
17	1	43	0	69	3	95	0	121	1
19	0	45	2	71	0	97	1	123	0
21	5	47	0	73	1	99	0	125	2
23	0	49	1	75	0	101	3	127	0
25	1	51	0	77	2	103	0	129	1

The pattern in $n(x)$ is a *ruler sequence.* This sequence seems to have two properties relevant to my observations: The sequence is symmetric about every local maxima and The sequence is self-similar under stripping.

Self Symmetry is portrayed through this table

number	n	**number**	n	**number**	n	**number**	n
8	3	152	3	296	3	440	3
32	5	176	4	320	6	464	4
56	3	200	3	344	3	488	3
80	4	224	5	368	4	512	9
104	3	248	3	392	3	536	3
128	7	272	4	416	5	560	4

The tabulation shows that each value of n owns an entire family of odd numbers. Listing the first few:

$$\begin{aligned} n = 0: &\quad 3, 7, 11, 15, 19, \ldots \\ n = 1: &\quad 1, 9, 17, 25, 33, \ldots \\ n = 2: &\quad 13, 29, 45, 61, \ldots \\ n = 3: &\quad 5, 37, 69, 101, \ldots \end{aligned}$$

Each of these is an arithmetic progression. For $n = 0$ the family steps by 4, For $n = 1$ family steps by 8, and so on. In general the **common difference** to a series corresponding to a certain value of n is:

$$d = 2^{n+2}$$

Upon collecting the **first term** of each series, for n from 0 to 9, A pattern is visible, though not immediately: the entries do not grow steadily, but *alternate* between two interleaved sequences.

n	**first term**
0	3
1	1
2	13
3	5
4	53
5	21
6	213
7	85
8	853
9	341

Taken separately, the even-n first terms are:

$$3, 13, 53, 213, 853, \ldots$$

and the odd-n first terms:

$$1, 5, 21, 85, 341, \ldots$$

Though these numbers don't make a clear arithmetic progression, the **differences** of two adjacent terms do **make a Geometric Progression.**

Using this pattern we derive the general pattern in first terms of series corresponding to each value of n.The first series produces:

$$T(r) = \frac{5 \cdot 4^r - 2}{6}, \qquad r = 1,2,3,\ldots$$

and the second series produces:

$$T(r) = \frac{4^r - 1}{3}, \qquad r = 1,2,3,\ldots$$

Now that we have a way to formulate the first term and the common difference of each arithmetic progression, we can find any term 'x' by using the general function:

$$x(n, r) = T(n) + (r - 1)\, 2^{n+2}$$

Thus the complete description of any odd number is the series, described by the number of halvings it will go through to reach the next odd number (i.e n) is:

$$x(n, r) = \begin{cases} \dfrac{2^{n+1} - 1}{3} + (r - 1)\, 2^{n+2}, & n \text{ odd} \\ \dfrac{5 \cdot 2^{n+2} - 2}{6} + (r - 1)\, 2^{n+2}, & n \text{ even} \end{cases}$$

$$\text{Halving count} : n \geq 0$$
$$\text{Position in the Series} : r \geq 1$$

The essential point is that $x(n, r)$ is a **bijection**. As n ranges over $\{0,1,2,\ldots\}$ and r over $\{1,2,3\ldots\}$, the values $x(n, r)$ run through every odd number once and only once, the series are **disjoint** (no odd number lies in two) and **exhaustive** (every odd number lies in one).

A map that hits every element of its target exactly once can be run backwards. Although $x(n, r)$ carries two variables, solving the series relation for r is $r = \dfrac{x - T(n)}{2^{n+2}} + 1,$ implying that r merely records where x sits inside its series, thus contributing nothing to which series x lies in.

The value of n is pinned down on its own, by a single requirement being $n(x)$ is the unique n for which the expression above yields a positive integer, equivalently, the unique n satisfying.

$$x \geq T(n) \quad \text{and} \quad x \equiv T(n) \pmod{2^{n+2}}$$

Since this partition is exact, only one n meets this condition, and no search beyond it is needed. The halving-count of x is thus already determined by the residue of x alone. Knowing where x falls among the series is equivalent to knowing $n(x)$

This derivation leads us to the **Final Result:**

$$Size = \sum_{i \in S} n\big(f_i(x)\big), \qquad S = \{\, i \geq 0 : f_i(x) \neq 1 \,\}$$

$$n(x) = \text{the unique } n \geq 0 \;\text{ with }\; x \equiv T(n) \pmod{2^{n+2}}$$

$$T(n) = \begin{cases} \dfrac{2^{n+1} - 1}{3}, & n \text{ odd} \\ \dfrac{5 \cdot 2^{n+2} - 2}{6}, & n \text{ even} \end{cases}$$

$$f(x) = \frac{3x + 1}{2^{\,n(x)+1}} \qquad f : 2\mathbb{Z} + 1 \to 2\mathbb{Z} + 1$$

$$f_i(x) = \underbrace{f(f(\cdots f(x)\cdots))}_{i \text{ times}}$$

Conclusion

This formulation takes any odd number and returns, without iteration, the exact structure of its next Collatz step, its family, indexed by its halving-count n, and its position within that family. Equivalently, it inverts the problem rather than computing $3x + 1$ and dividing down to find $n(x)$ it reads $n(x)$ directly off the residue of x. Every odd number is placed in exactly one arithmetic progression, and the progressions are given in closed form.

However, this method is limited because the construction determines a single step completely, but says nothing about how the halving-count evolves as the map is applied along a full trajectory. This is the exact point at which Collatz becomes hard.

The method here is unique, we only use elementary mathematics like arithmetic progressions and a finite-difference argument. It reaches the residue structure of the odd numbers by a direct, self-contained route. The map f is the classically studied *Syracuse function*, and the residue behaviour of Collatz trajectories has been examined before, notably by Terras (1976) and Everett (1977) and surveyed by Lagarias (1985). The present treatment arrives at the single-step structure independently and by different means, and frames the halving-counts through the *ruler sequence* a lens that makes the self-similarity immediate and the residue structure lucid .